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ScienceDirect
Proceedings of the Combustion Institute 33 (2011) 477–484
Proceedings of the Combustion Institute
www.elsevier.com/locate/proci

Elementary kinetics for gas phase combustion of $SiCl_4$ based mixtures

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Available online 15 September 2010

Abstract

An experimental study has been performed to examine some elementary reactions of the complex kinetics of $SiCl_4$ based mixtures used for the formation of silica (SiO_2) synthesized by gas phase combustion. The reactivity of $SiCl_4/N_2O$ mixtures was studied behind reflected shock waves and the atomic resonance absorption spectroscopy (ARAS) was applied for time-resolved measurements of O-atoms in gas mixtures, highly diluted in argon, containing up to 100 ppm $SiCl_4$ and up to 20 ppm N_2O. The absorption profiles obtained are kinetically evaluated by a modeling study based on a detailed reaction mechanism with 61 species and 89 reactions. Rate constants for reactions

$$SiCl_3 + O \rightarrow \text{products}$$
$$SiCl_2 + O \rightarrow SiClO + Cl$$

are determined at temperatures comprised between 1750 and 3010 K and pressures comprised between 1.1 and 2.4 bar.
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Keywords: $SiCl_4$; $SiCl_3$; $SiCl_2$; Kinetics; Combustion

1. Introduction

Kinetic data for the modeling of processes dealing with the production of materials via gas phase combustion synthesis are needed [1,2]. Unfortunately, the gas phase kinetics of silicon containing species is rarely studied, especially at high temperatures. Therefore, the detailed reaction pathways for the formation of SiO_2 particles by gas phase combustion of silicon tetrachloride ($SiCl_4$) based mixtures are generally not understood and for a lot of them no kinetic data are available. Concerning the elementary kinetics of $SiCl_4$ and radicals formed from silicon tetrachloride some fundamental works have been performed at high temperatures, behind a reflected shock wave, by applying the ARAS technique [3–9]. However, these works do not allow the modeling of experiments with SiO_2 particles production via gaseous mixtures such as $SiCl_4/CH_4/O_2$ (Ulrich and Riehl

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doi:10.1016/j.proci.2010.05.009

[10], Allendorf et al. [11]), $SiCl_4/H_2/O_2$ (Katz and Hung [12], Chung et al. [13], Hung and Katz [14], Hannebauer and Menzel [15]) or $SiCl_4/O_2$ (French et al. [16], Okuyama et al. [17], Binnewies and Jug [18]). Although various effects of reactor conditions on the formation kinetics of silica particles have been observed and reported in the above-cited works and references therein, the fundamental gas-phase and surface chemistry occurring in these systems remains largely unstudied. Recently, Moore et al. [19] proposed a detailed chemical kinetic model for the modeling of $SiCl_4/H_2/O_2$ flames. These authors adjusted six rate constants, for which previous estimates or measurements were available, to get a good agreement between modeling and experiments. This strategy for such a less-well understood system is appropriate. However, the combustion of the H_2/O_2 system forms O atoms through the chain-branching reaction

$$H + O_2 \rightarrow OH + O$$

and other reactions. Therefore, reactions between O atoms and $SiCl_4$ and radicals such as $SiCl_3$ are feasible but not considered in the $SiCl_4/H_2/O_2$ detailed kinetic model proposed by Moore et al. [19]. Thus, it is possible that the fair agreement between experiments and calculations with the unadjusted model is caused by the lack of some important reactions and information on whether they can even occur. This remains to be demonstrated and is beyond the scope of the present paper. Hannebauer and Menzel [15] did not consider these $SiCl_x + O$ reactions although they recognize that $SiCl_4$ could react with any of the intermediates occurring during H_2/O_2 reactions. In the present investigation, $SiCl_4/N_2O/Ar$ mixtures are used to study the reaction of O atoms with $SiCl_4$ and derived radicals, mostly $SiCl_3$ and $SiCl_2$, by detecting O atom profiles at 130.5 nm.

2. Experimental

The experiments were carried out in a stainless steel pressure driven shock tube with an internal diameter of 79 mm. It is divided by a thin aluminum diaphragm into a driver section of 3.5 m and a driven section of 5.7 m in length. Gases used in the present study were of high purity, i.e. Ar > 99.9999 mol% and N_2O > 99.97 mol%. $SiCl_4$ (purity > 99.999 mol%) is a liquid at room temperature. It was injected and evaporated in a separate stainless steel vessel. The shock tube is equipped with diagnostics for atomic resonance absorption spectroscopy (ARAS) consisting of a microwave excited discharge lamp. The lamp was operated with a flowing gas mixture containing N_2O in He. A series of carefully controlled shock wave calibrations have been done to relate the measured absorptions to the corresponding O atom concentrations. The O atom calibration is based on the thermal decomposition of N_2O highly diluted in Ar above 1800 K. In such conditions, the O atom absorption increases gradually to reach a plateau. The time-dependent absorption of O atoms $A(t)$ is given by

$$A(t) = 1 - \exp\{-\ell\sigma_O[O(t)]^n\}$$

where ℓ is the optical path length, σ_O the cross section of O atoms and $[O(t)]$ is the time-dependent oxygen concentration. The calibration curve of the O atoms is expressed by a modified Lambert–Beer expression. This modification represents a calibration of the experimental system.

3. Results and discussion

The fate of O atoms in $SiCl_4/N_2O$ mixtures highly diluted in argon has been investigated in the 1750–3000 K temperature range. Briefly, the O atoms signal increases rapidly, through N_2O thermal decomposition, peaks and then decreases. This behavior was observed in all $SiCl_4/N_2O$ experiments. Typical examples are shown Figs. 1 and 2. In N_2O/Ar mixtures it is well known, as observed here in calibration experiments, that the O atoms concentration increases gradually to reach a plateau. Since both oxygen atoms and $SiCl_4$ are absorbing light at 130.5 nm the time-dependent absorption, hereafter noted $A(t)$ is:

$$\ln\ (1 - A(t)) = -\ell(\sigma_O[O(t)]^n + \sigma[SiCl_4(t)])$$

where $[O(t)]$ represents the time-dependent concentration in O atoms, $[SiCl_4(t)]$ the time-dependent concentration in $SiCl_4$ molecules, σ the absorption cross-section of $SiCl_4$ at 130.5 nm, σ_o the absorption cross-section of O atoms at 130.5 nm and ℓ the optical path length. The

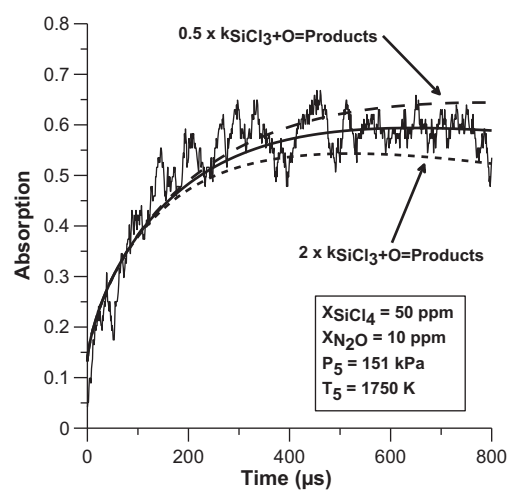

Fig. 1. Comparison between measured and calculated (solid line) O atom absorption profiles with variation of k_1 and k_2 (dashed lines).

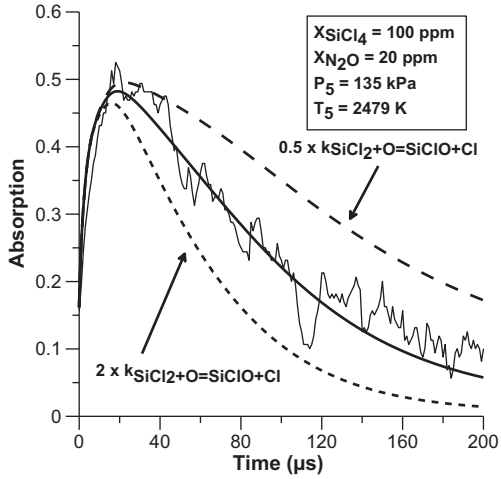

Fig. 2. Comparison between measured and calculated (solid line) O atom absorption profiles with variation of k_6 (dashed lines).

absorption cross-section σ was measured to be 5.7×10^{-17} cm^2 mol^{-1}. The absorption due to N_2O is negligible since the absorption cross-section σ_{N2O} is 6.8×10^{-18} cm^2 mol^{-1}.

A detailed kinetic model has been built. It consists of 89 reactions and 61 species. This detailed chemical kinetic model and related thermochemistry are given as Supplemental data. The Si/O/Cl sub-mechanism of the Si/O/Cl/H detailed kinetic model proposed by Moore et al. [19] is considered, some reactions are discarded or updated, some other added to interpret the present experimental results such as $SiCl_3 + O \rightarrow$ Products, $SiCl_2 + O \rightarrow$ Products and some Si/O reactions and some Si/N and Si/N/O reactions. For each experiment, the time-dependent $SiCl_4$ absorption and the time-dependent O atoms absorption have been estimated from the kinetic model. It is thus possible to compare the absorption experimental profiles and the absorption computed profile. Four series of mixtures have been studied, one with 50 ppm $SiCl_4$, one with 100 ppm $SiCl_4$, one with 150 ppm and the last one contained 200 ppm $SiCl_4$. At $T > 2000$ K, the absorption signal never returns to zero indicating that some reaction products absorb the light. This is markedly true for the mixtures containing initially 150 and 200 ppm $SiCl_4$. Since the absorbing species are unknown and their corresponding absorption cross-section as well, it is more appropriate to consider only the experiments containing initially up to 100 ppm $SiCl_4$.

An attempt to explain the O-atom time histories by the reaction

$$SiCl_4 + O \rightarrow SiCl_3 + ClO$$

was not successful even with unrealistically high, i.e. above the gas kinetic limit, rate coefficients. This is not surprising because this channel is 194 kJ mol^{-1} endothermic. In fact, all the

$$SiCl_4 + O \rightarrow \text{products}$$

channels, whatever the products are, endothermic by several hundreds of kJ mol^{-1}. Therefore, the O atom profiles have to be explained by reactions between O atoms and radicals, namely $SiCl_3$, $SiCl_2$ and SiCl or by reactions between O atoms and Si atoms. Allendorf and Melius [20] supposed that $SiCl_4$ oxidation is initiated by $SiCl_4 \rightarrow SiCl_3 + Cl$ and $SiCl_3 \rightarrow SiCl_2 + Cl$ and that species likely to react are $SiCl_3$ and $SiCl_2$. These suppositions are confirmed here. They discussed numerous possible reaction pathways for $SiCl_3 + O_2$ and $SiCl_2 + O_2$ but did not discuss $SiCl_3 + O$ and $SiCl_2 + O$ reactions. The oxidation of $SiCl_3$ can proceed via product channels:

$$^2SiCl_3 + {}^3O \rightarrow {}^1SiCl_2 + {}^2ClO \qquad \Delta_{rxn}H^\circ = +3.46 \text{ kJ mol}^{-1} \tag{R1}$$

$$^2SiCl_3 + {}^3O \rightarrow {}^1SiCl_2O + {}^2Cl \qquad \Delta_{rxn}H^\circ = -276 \text{ kJ mol}^{-1} \tag{R2}$$

with the more probable exothermic channel (R2). Formation of $SiCl_2O$ in $SiCl_4/O_2$ mixtures has been assessed by Junker et al. [21]. Thermodynamic data for $SiCl_2O$ are available in Allendorf and Melius [20]. It exists controversial data for the enthalpy of formation of $SiCl_3$ [22,23]. The enthalpy of formation given by Su and Schlegel [23] is considered in this study. The formation of $SiCl_3O$ by

$$SiCl_3 + O + M \rightarrow SiCl_3O + M$$

is possible but such reactions are generally negligible in the pressure range considered here (around 150 kPa). The replacements of (R1) and (R2) by the following reactions (all the other reactions remaining unchanged):

$$SiCl_3 + O + M \rightarrow SiCl_3O + M$$

$$SiCl_3O + M \rightarrow SiCl_2O + Cl + M$$

do not allow the interpretation of our experimental data, whatever the rate constants are, and it can be concluded that no significant amount of $SiCl_3O$ is formed at such experimental conditions. The same holds when these two reactions are added to the full kinetic model (R1) and (R2) included).

The occurrences of reactions between $SiCl_3$ or $SiCl_2$ with N_2O, the O atoms precursor, need to be examined. Golberg et al. [24] studied the reaction $SiCl_2 + N_2O$ in the 600–800 K temperature range and observed $SiCl_2O$ and N_2 as products but did not report any rate constant. Safarik et al. [25] reported a rate coefficient of 5.7×10^{11} cm^3 mol^{-1} s^{-1} at 298 K for $SiCl_2 + N_2O \rightarrow$ products. Becerra and Walsh [26] report a rate constant of 4.89×10^{11} exp(+2000 J/RT) cm^3 mol^{-1} s^{-1} in

the 295–747 K temperature range for $SiH_2 + N_2O$. The reaction has a negative temperature dependence with an activation energy of – 2000 J mol^{-1} with a rate coefficient of 1.14×10^{12} cm^3 mol^{-1} s^{-1} at 298 K. The rate constant for $SiCl_2$ is lower than the rate constant measured for SiH_2. According to Becerra and Walsh [26], where comparisons are possible SiH_2 is generally found more reactive than $SiCl_2$. Therefore, by analogy with $SiH_2 + N_2O \rightarrow SiH_2O + N_2$ and its negative temperature dependence, an upper limit of 5.7×10^{11} cm^3 mol^{-1} s^{-1} is considered for $SiCl_2 + N_2O \rightarrow SiCl_2O + N_2$ at high temperature. No data is reported for $SiCl_3 + N_2O$ reaction. Slagle et al. [27] report an upper limit of 3×10^9 cm^3 mol^{-1} s^{-1} at 500 K for $SiH_3 + N_2O \rightarrow$ other products $+ N_2$. This represents a much lower rate constant for $SiH_3 + N_2O$ that the one observed for $SiH_2 + N_2O$ and therefore, it can be assumed that the rate constant for the $SiCl_3 + N_2O$ reaction is much lower than the rate constant for $SiCl_2 + N_2O$ reaction. However, even the consideration of a rate constant of 5.7×10^{11} cm^3 mol^{-1} s^{-1} for $SiCl_3 + N_2O \rightarrow SiCl_3O + N_2$ and for $SiCl_2 + N_2O \rightarrow SiCl_2O + N_2$ does not explain the O atoms profiles observed and these reactions appear to play a marginal role whatever the temperature is in the 1750–3010 K temperature range. Attempts to explain the experimental profiles by only one R1 or R2 of the above reactions or both (R1 + R2) reactions were not successful in all cases. Therefore, reactions concerning $SiCl_2$ have to be added. The reactions under consideration are:

$$^1SiCl_2 + {}^3O \rightarrow {}^2SiCl + {}^2ClO$$
$$\Delta_{rxn}H^\circ = +179.0 \text{ kJ mol}^{-1} \quad \text{(R3)}$$
$$^1SiCl_2 + {}^3O \rightarrow {}^1SiO + {}^2Cl + {}^2Cl$$
$$\Delta_{rxn}H^\circ = +61.8 \text{ kJ mol}^{-1} \quad \text{(R4)}$$
$$^1SiCl_2 + {}^3O \rightarrow {}^1SiO + {}^1Cl_2$$
$$\Delta_{rxn}H^\circ = -180.7 \text{ kJ mol}^{-1} \quad \text{(R5)}$$
$$^1SiCl_2 + {}^3O \rightarrow {}^2SiClO + {}^2Cl$$
$$\Delta_{rxn}H^\circ = -120 \text{ kJ mol}^{-1} \quad \text{(R6)}$$

Reaction (R3) is too endothermic to be of some importance here and the same holds for reaction (R4). Reaction (R5) is spin-forbidden since the ground-state $SiCl_2$ is a singlet. Therefore, this reaction will exist with a negligible rate constant compared to (R6), which is spin allowed. Rate constant for the thermal decomposition of $SiCl_4$

$$SiCl_4 + M \rightarrow SiCl_3 + Cl + M$$
$$SiCl_3 + M \rightarrow SiCl_2 + Cl + M$$
$$SiCl_2 + M \rightarrow SiCl + Cl + M$$
$$SiCl + M \rightarrow Si + Cl + M$$

are taken in Kunz and Roth [5]. Concerning the initiation reaction of silicon tetrachloride thermal decomposition, neither experimental nor theoretical works support that the initiation reaction is

$$SiCl_4 + M \rightarrow SiCl_2 + Cl_2 + M$$

instead of

$$SiCl_4 + M \rightarrow SiCl_3 + Cl + M.$$

Kunz and Roth [5] followed the formation and fate of Cl atoms during $SiCl_4$ thermal decomposition and explained their observations by $SiCl_4 + M \rightarrow SiCl_3 + Cl + M$. Theoretical study by Su and Schlegel [23] also supports this conclusion.

Other reactions such as

$$SiCl_3 + SiCl_3 \rightarrow SiCl_4 + SiCl_2$$
$$Si + SiCl_4 \rightarrow SiCl_2 + SiCl_2$$

potentially of importance here are reported as given by Moore et al. [19] and by Kunz and Roth [3], respectively. Other reactions relevant to the thermal decomposition of silicon tetrachloride are included in the detailed chemical kinetic model. The agreement of this mechanism with the thermal decomposition of $SiCl_4$ is presented in Fig. 3.

The evaluations of

$$SiCl_3 + O \rightarrow \text{products}$$

and

$$SiCl_2 + O \rightarrow SiClO + Cl$$

are performed by fitting calculated absorption profiles to the measured ones by variations of (R1), (R2) and (R6). It was assumed that both channels (R1) and (R2) have the same rate coefficients. In Figs. 1 and 2 the results of the fitting procedure is presented as a solid line.

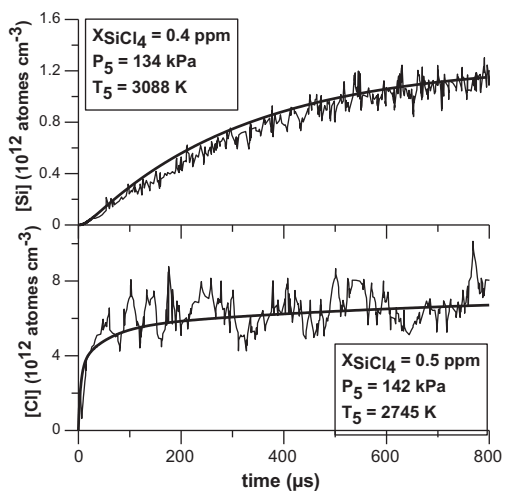

Fig. 3. Comparison between experimental and calculated (solid lines) Cl atom and Si atom profiles during the thermal decomposition of $SiCl_4$ in argon. Experimental data are from Kunz and Roth [7].

The sensitivity of the rate coefficients k_1 and k_2 is shown in Fig. 1 by variations by factors 0.5 and 2 (see dashed lines in Fig. 1). The sensitivity of the rate coefficient k_6 is shown in Fig. 2 by variations by factors 0.5 and 2 (see dashed lines in Fig. 2). These brute force sensitivity analyses, although appropriate, need to be completed by first-order sensitivity analyses. These analyses are presented Figs. 4 and 5 for experimental conditions presented Figs. 1 and 2, respectively. The normalized sensitivity to the formation of O atoms is defined as $\frac{k_i}{[O]}.\frac{d[O]}{dk_i}$. It can be seen from these figures that the reactions influencing the formation of O atoms depend on the reaction times and on the temperature. For the lowest temperatures in this study (about 1750 K) the most influential reactions are, as expected, the thermal decomposition of N_2O and the initiation step of the thermal decomposition of $SiCl_4$ as shown in Fig. 4. The thermal decomposition of N_2O can be considered as well known since this system was studied exhaustively. The rate constant $SiCl_4 + M \rightarrow SiCl_3 + Cl + M$ was studied [6,8] and is known with a $\Delta k/k$ of 50% at best. The rate constant given by Kunz and Roth in the 1560–2200 K temperature range around 1.6 bar is extrapolated at higher temperature. Reactions (R1) and (R2) are then the most influential with reaction (R1) being the most sensitive at reaction times longer than 400 μs. Also sensitive, but much less than (R1) and (R2), are $SiCl_3 + M \rightarrow SiCl_2 + Cl + M$ [5,8] and $O_2 + Cl \rightarrow ClO + O$. Figures 4 and 6 show that reaction (R6) is very weakly sensitive at these conditions and it is possible to derive rate coefficients for (R1) and (R2) almost unambiguously.

At temperature about 2400 K, as shown Fig. 5, the thermal decomposition of N_2O remains very sensitive whereas the initiation step of $SiCl_4$ thermal decomposition does not appear sensitive any-

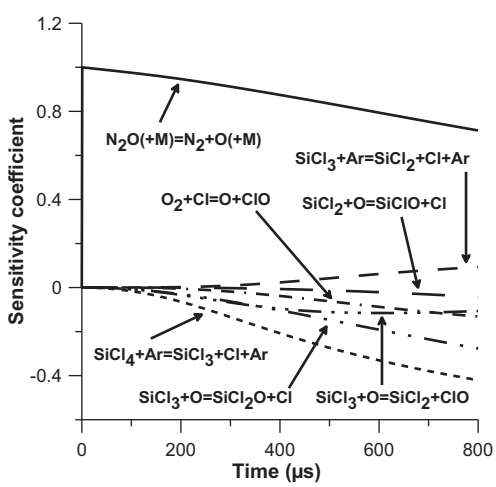

Fig. 4. Normalized sensitivity to O atoms. Sensitivities are calculated for the conditions of Fig. 1.

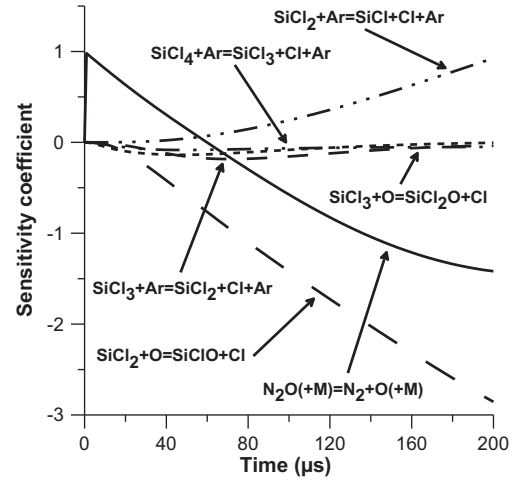

Fig. 5. Normalized sensitivity to O atoms. Sensitivities are calculated for the conditions of Fig. 2.

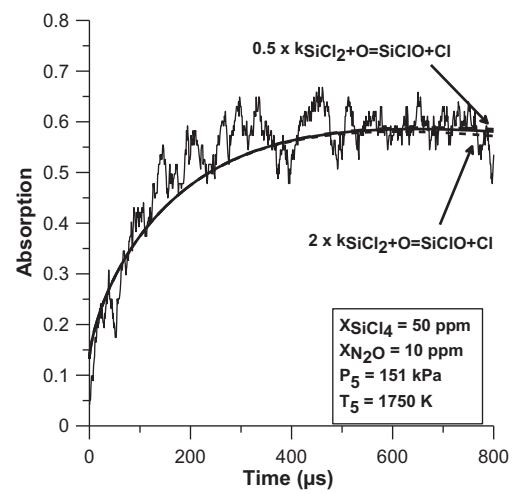

Fig. 6. Comparison between measured and calculated (solid line) O atom absorption profiles with variation of k_6 (dashed lines).

more. Interestingly, reactions (R1) and (R2) are sensitive for short reaction times (here up to about 20 μs) and the most sensitive reaction at longer reaction times is (R6). Figure 7 shows that (R1) and (R2) are of importance up to the peak. $SiCl_3 + M \rightarrow SiCl_2 + Cl + M$ is not sensitive whereas $SiCl_2 + M \rightarrow SiCl + Cl + M$ is increasingly sensitive with time. This last reaction has been studied [5,8] and is known with a $\Delta k/k$ of 50% at best. It is therefore, possible to derive from these experiments the rate coefficients of $SiCl_2 + O \rightarrow SiClO + Cl$ almost unambiguously. At intermediate temperature between 1800 and 2400 K, reactions (R1), (R2) (mostly up to the peak) and (R6)

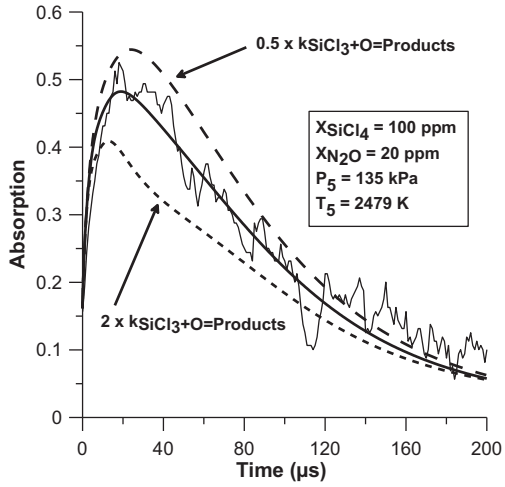

Fig. 7. Comparison between measured and calculated (solid line) O atom absorption profiles with variation of k_1 and k_2 (dashed lines).

(mostly after the peak), together with $SiCl_4 + M \rightarrow SiCl_3 + Cl + M$, $SiCl_3 + M \rightarrow SiCl_2 + Cl + M$ and $SiCl_2 + M \rightarrow SiCl + Cl + M$, are responsible for the O atom absorption profiles measured. By variation of the rate coefficients of reactions (R1), (R2) (with $k_1 = k_2$) and (R6), best fits to the measured profiles were tried to achieve. As shown in Figs. 8 and 9, this procedure leads to satisfactory results. All experiments were evaluated in the manner just described. The best fit values of k_1, k_2 and k_6 obtained are summarized in the Arrhenius diagram of Fig. 10. The data points can be well represented by the following modified Arrhenius expressions:

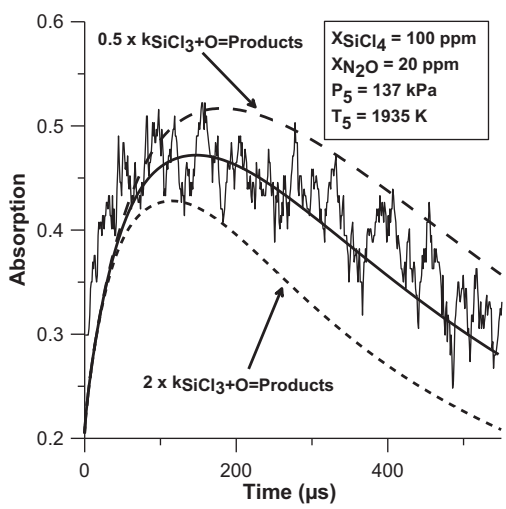

Fig. 8. Comparison between measured and calculated (solid line) O atom absorption profiles with variation of k_1 and k_2 (dashed lines).

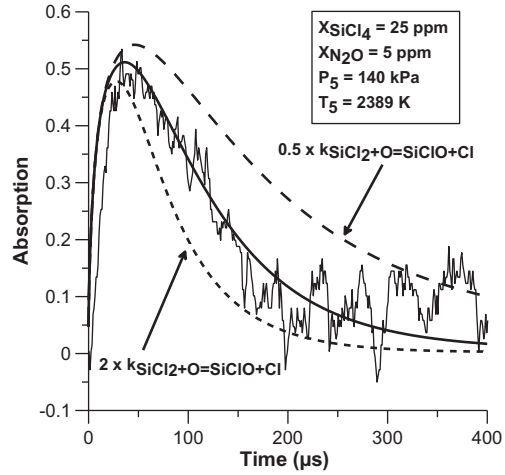

Fig. 9. Comparison between measured and calculated (solid line) O atom absorption profiles with variation of k_6 (dashed lines).

$$k_1 = k_2\left(\mathrm{cm^3\,mol^{-1}\,s^{-1}}\right) = 5 \times 10^{13} T^{0.8} \exp\left(\frac{-106.2\,\mathrm{kJ}}{\mathrm{RT}}\right)$$

$$k_6\left(\mathrm{cm^3\,mol^{-1}\,s^{-1}}\right) = 10^{8} T^{2.2} \exp\left(\frac{-70.8\mathrm{kJ}}{\mathrm{RT}}\right)$$

These equations represent the best-fit. It is quite difficult to compare these rate constants with data in the literature because no data are available for $SiCl_3 + O \rightarrow$ products and $SiCl_2 + O \rightarrow$ products. Taylor et al. [28] report a rate coefficient of 8.4×10^{13} cm^3 mol^{-1} s^{-1} at 298 K for $SiH_3 + O \rightarrow$ Products and Ryan and Plumb [29] report a rate coefficient of 2.5×10^{13} cm^3 mol^{-1} s^{-1} at 295 K for $CCl_3 + O \rightarrow CCl_2O + Cl$. $SiCl_3$ being less reactive than CCl_3 and less reactive than SiH_3, rate

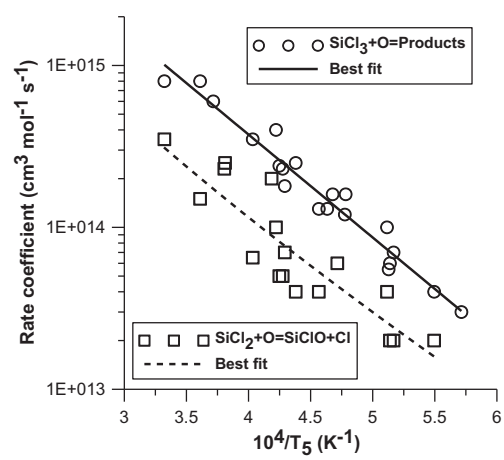

Fig. 10. Arrhenius representation of $k_1 + k_2$ (k_1 for $SiCl_3 + O \rightarrow SiCl_2 + ClO$, k_2 for $SiCl_3 + O \rightarrow SiCl_2O + Cl$ with $k_1 = k_2$) and k_6 for $SiCl_2 + O \rightarrow SiClO + Cl$.

coefficients of the order 10^{13} up to 10^{14} cm^3 mol^{-1} s^{-1} observed for $SiH_3 + O$ and $CCl_3 + O$ at about 298 K are observed at much higher temperature for $SiCl_3 + O$.

No rate constant, even at room temperature, is available in the literature for $SiCl_2 + O$ and the corresponding reactions $SiH_2 + O$ and $CCl_2 + O$. Theoretical calculations have been performed for the reaction $SiCl_2 + H \rightarrow SiCl + HCl$ in the 300–2000 K temperature range and the activation energy for this reaction is found to be equal to 78.9 kJ mol^{-1}. The activation energies for these two reactions are found to be similar for the same reaction type, i.e. a Cl abstraction from $SiCl_2$ by an atom.

The uncertainty of these rate coefficients is relatively high, $\Delta k/k = 50\%$, as a result of all the uncertainties, including the uncertainties of N_2O thermal decomposition, of $SiCl_4$ thermal decomposition, of temperature and of the absorption cross section of silicon tetrachloride. For k_1 and k_2, the uncertainty is higher for the highest temperatures whereas for k_6 the uncertainty is higher for the lowest temperatures.

4. Conclusions

The reactivity of $SiCl_3$ and $SiCl_2$ radicals with O atoms has been investigated by measuring O-atom absorption behind reflected shock waves in $SiCl_4/N_2O$ mixtures highly diluted in argon at temperatures between 1750 and 3010 K and pressures between 1 and 2.4 bar. These reactions have not received a lot of attention in the literature. A detailed chemical kinetic model has been built to examine the corresponding gas-phase chemistry. This model is able to predict all measured O-atom profiles and rate constants are derived. The results are needed for the understanding and modeling of silica particle production during the gas-phase combustion synthesis of silica particles. The $SiCl_4/N_2O$ mixture is not under consideration for combustion synthesis but the $SiCl_x + O$ reactions observed are likely to occur in $SiCl_4/H_2/O_2$ mixture, which is used for combustion synthesis of silica particles. The fact that these reactions are important or not compared to $SiCl_x + H$ reactions or to $SiCl_x + OH$ reactions is not discussed in this study and these interesting issues remain to be studied. The results can also be of some uses for the modeling of thermal chemical vapor deposition (CVD) processes. Complementary experimental studies as well as the reappraisals of previous studies are needed. Few experimental studies dealing with the elementary kinetics of silicon compounds are performed, especially with chlorinated silicon compounds, and each new study can induce potential changes in the kinetic interpretation of some observed trends. Theoretical studies by using the transition state theory (TST) are also needed but their ability to be predictive for the elementary kinetics of silicon species remains to be demonstrated more thoroughly as it is nowadays.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.009.

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